



# Soybean yield prediction by machine learning and climate

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## Abstract

Soybean cultivation plays an important role in Mato Grosso do Sul and around the world. Given the inherent complexity of the agricultural system, this study aimed to develop climate-based yield prediction models using ML, considering the most correlated meteorological variables for each condition, test the best model with independent data, and define zones of higher soybean yield in Mato Grosso do Sul to recommend better planting sites. The study was carried out in two stages. First, meteorological and soybean yield data obtained from 47 locations in the state of Mato Grosso do Sul were used to calibrate the machine learning (ML) algorithms. Second, the best algorithm was used to predict soybean yields throughout Mato Grosso do Sul. Daily meteorological data of air temperature ( $T$ , °C), precipitation ( $P$ , mm), global solar irradiance ( $Q_g$ , MJ m<sup>-2</sup> day<sup>-1</sup>), wind speed ( $u_2$ , m s<sup>-1</sup>), net radiation ( $R_n$ , MJ m<sup>-2</sup> day<sup>-1</sup>), and relative humidity ( $RH$ , %) of the NASA-POWER system from 2002 to 2021 were used. The reference evapotranspiration ( $ETo$ ) by the standard FAO method and water balance (WB) by Thornthwaite and Mather (1955) were calculated for each collection point. The MLs used in this stage consisted of multiple linear regression (MLR), multilayer perceptron (MLP), support vector machine (SVM), random forest (RF), extreme gradient boosting (XGBOOSTING), and gradient boosted decision (GradBOOSTING). The ML models were calibrated using 70% of the data selected for training and 30% for validation. Algorithms were evaluated by accuracy, precision, and tendency. All analyses were performed using Python 3.8 software. Climate variables showed high spatial and seasonal variability throughout Mato Grosso do Sul (MS). Pearson's univariate correlations between soybean yield and climate variables of the phenological period showed distinct relationships and different intensities. For instance, soil water storage (ARM) showed negative, neutral, and positive correlations in October, November, and December, respectively. The calibrated ML algorithms had a high precision and accuracy in both calibration and testing. For instance, the best model in the calibration was XGBOOSTING, which showed MAPE,  $R^2$ , RMSE, MSE, and MAE values of 1.84%, 0.95, 2.06%, 4.24%, and 0.921%, respectively. Random forest (RF), extreme gradient boosting (XGBOOSTING), and gradient boosting (GradBOOSTING) were the most precise machine learning algorithms, with  $R^2$  values of 0.71, 0.62, and 0.62 in the test, respectively.

## 1 Introduction

Advance yield predictions assist in decision-making at all levels of the production chain more efficiently (Cardoso et al. 2010; Sarker et al. 2019), allowing the estimation of the investment level, collaboration to regulate prices, reduced speculation, and increased profitability and sustainability for producers (Sørensen et al. 2019).

The agricultural system is an example of an extremely complex system (Camargo et al. 1998), in which traditional crop modeling, which includes agrometeorological (L. E. Aparecido et al. 2016; Victorino et al. 2016), agrometeorological-spectral (Rosa et al. 2010; Volpato 2013), mechanistic (Santos and Camargo 2006), and phytotechnical models (Miranda et al. 2014) often fail to accurately estimate

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crop yield, as these models do not capture all the necessary complexity (Chan et al. 2020; Isabella and Srinivasan 2018; Mahajan et al. 2016).

The data volumes, which are produced by automatic weather stations, generally referred to as big data, require gigantic storage resources, in addition to editing, analysis, and interpretation (Cravero and Sepúlveda 2021). The latter has enormous potential to add value to society, the environment, and decision-makers (Sonka 2016). However, the interpretation of these data involves challenges due to their volume, variety, speed, veracity, and value (Meng et al. 2020). Conventional data processing techniques are not able to meet the constant demands in the growth of smart agriculture, becoming a major obstacle to extracting valuable information from field data (Evstatiev and Gabrovska-Evstatieva 2021). However, the emergence of Machine Learning (ML), which is a subset of artificial intelligence, has been used in the exponential growth of computational potential capacity (Helm et al. 2020). According to Benos et al. (2021), machine learning (ML) is the vanguard in modeling complex systems in the world, being part of the study area of artificial intelligence (AI). ML was defined by Samuel (1959) and Mitchell (1997) as autonomous interactive mathematical processes, that is, computers act without explicit programming for the solution of a given problem. ML allows computers to learn from errors in performance metrics (Gao et al. 2018). ML comprises a large set of types of analysis that seek to find patterns, both univariate and multivariate, performing optimizations, classifications, and estimations/predictions.

ANNs are types of nonlinear modeling inspired by the functioning of a biological brain, being fundamentally capable of finding patterns from available data (Sun and Cheng 2012). They learn from experience and can estimate any complex functional relationship with great accuracy. ANNs have an input layer, a variable number of hidden layers, and an output layer, all composed of neurons. Neurons are the basic calculation units of an ANN (Rabunal and Dorado 2006).

Typically, traditional cropping models deal with the problem of development and production of a plant community in a multivariate way from univariate and/or bivariate sub-models, as in the case of the DSSAT (Hoogenboom 1997) and AQUACROP system models (Vanuytrecht et al. 2014). For instance, air temperature is used to estimate the duration of each phase and even the total duration of a crop cycle using the thermal index concept (Santos and Camargo 2006). However, this duration can be changed due to several factors together (Al-Jarrah et al. 2015; Sassi et al. 2019), which are not directly accounted for, such as water deficit, photoperiod, evapotranspiration, and relative humidity, depending on the moment and intensity of occurrence of these factors (Ferreira et al. 2019). MLs are adapted to handle these types of

problems as long as the amount of data is large enough for the MLs to 'learn' the patterns (Che et al. 2013; Chen and Xiaotong 2014).

Big data and ML have shown promising advances in agricultural engineering (Slavakis et al. 2014) and have brought with them enormous challenges, but research in this area is revolutionary (Bhatnagar 2018; Qiu et al. 2016). A study by McKinsey Global Institute showed that ML will play a notable role in the big data revolution (James and Chui 2011). This study shows the ability of ML to learn from data and provide insights, decisions, and predictions (Neethirajan 2020). Several methodologies can be used to overcome these challenges, including a series of adaptations for the correct use of ML in Big Data systems, mainly in agriculture (L'Heureux et al. 2017).

Climate is one of the main factors causing the decrease in soybean yield (Sentelhas et al. 2015), and water deficit is one of the elements that most limit production among the climate variables (Battisti et al. 2017), with soil water condition being a sensitive indicator to the expression in grain yield (Martorano et al. 2009). Câmara (1991) also reported that the reduction in soybean growth, development, and production is due to the variation in meteorological factors in a growing region.

Soybean can present a full productive potential under suitable climate conditions and, consequently, express an increased yield as long as there are no other limiting factors (Smiderle et al. 2019). Among the meteorological elements, air temperature, solar radiation, soil moisture, and water deficit are determinants of the efficiency of physiological processes in plants (Battisti et al. 2017).

Many studies have dealt with the relationship between climate and soybean yield, but no national or international publication has applied knowledge of MLs to predict yield based on climate conditions. In this sense, this study hypothesized that soybean yield estimation with machine learning algorithms might be a tool to assist in searching for strategies that minimize the losses resulting from variability and climate change.

In the world, some productivity forecasting works using artificial intelligence are found. For example, Jin et al. (2017) who estimated corn productivity for the USA. Chen et al. (2022) which predicts corn yield in China. However, no studies were found that predicted soybean yield using AI climate.

Crop forecasts are currently based mainly on interviews and subjective surveys. These are low-yield ways, in addition to being expensive and with low accuracy. Thus, given the importance of soybean cultivation in Mato Grosso do Sul and the inherent complexity of the agricultural system, this study aims to develop climate-based yield prediction models using ML, considering the most correlated meteorological variables for each condition, test the best model with

independent data, and define zones of higher soybean yield in Mato Grosso do Sul to recommend better planting sites.

## 2 Methodology

### 2.1 Meteorological data and soybean yield

The study was carried out in two stages. First, meteorological and soybean yield data obtained from 47 locations in the state of Mato Grosso do Sul were used (Fig. 1) to calibrate the machine learning (ML) algorithms. The yield data were provided by cooperatives and producers in the region. In the second stage, the most accurate and precise ML algorithm was used to be applied throughout the state of Mato Grosso do Sul.

### 2.2 1st stage—ML calibration

Daily meteorological data of air temperature ( $T$ , °C), precipitation ( $P$ , mm), global solar irradiance ( $Q_g$ ,  $\text{MJ m}^{-2} \text{ day}^{-1}$ ), wind speed ( $u_2$ ,  $\text{m s}^{-1}$ ), net radiation ( $R_n$ ,  $\text{MJ m}^{-2} \text{ day}^{-1}$ ),

and relative humidity ( $RH$ , %) of the NASA-POWER system from 2002 to 2021 were used. The data were collected at each geographic coordinate referring to each point of soybean yield provided by the cooperative. The NASA-POWER grid is  $0.5^\circ \times 0.5^\circ$  (approximately  $56 \times 56 \text{ km}$ ) and the data comes from various global surface measurements and remote sensing systems and meteorological modeling.

The meteorological data were stratified every 10 days, with total values used for precipitation and mean values for the other meteorological elements. The NASA-POWER system has good data quality for Brazil, with good accuracy relative to surface data, especially in 10-day scale data (Monteiro et al. 2018). The reference evapotranspiration ( $ET_o$ ) was calculated for each collection point by the FAO standard method, proposed by Penman–Monteith (Allan et al. 1998), considering the sensible heat flux in the soil equal to 5% of the net radiation. The water balance ( $WB$ ) was calculated according to Thornthwaite and Mather (1955), in which the following elements were obtained:  $ET_r$  is the real evapotranspiration (mm),  $ARM$  is the soil water storage (mm),  $DEF$  is the water deficit of

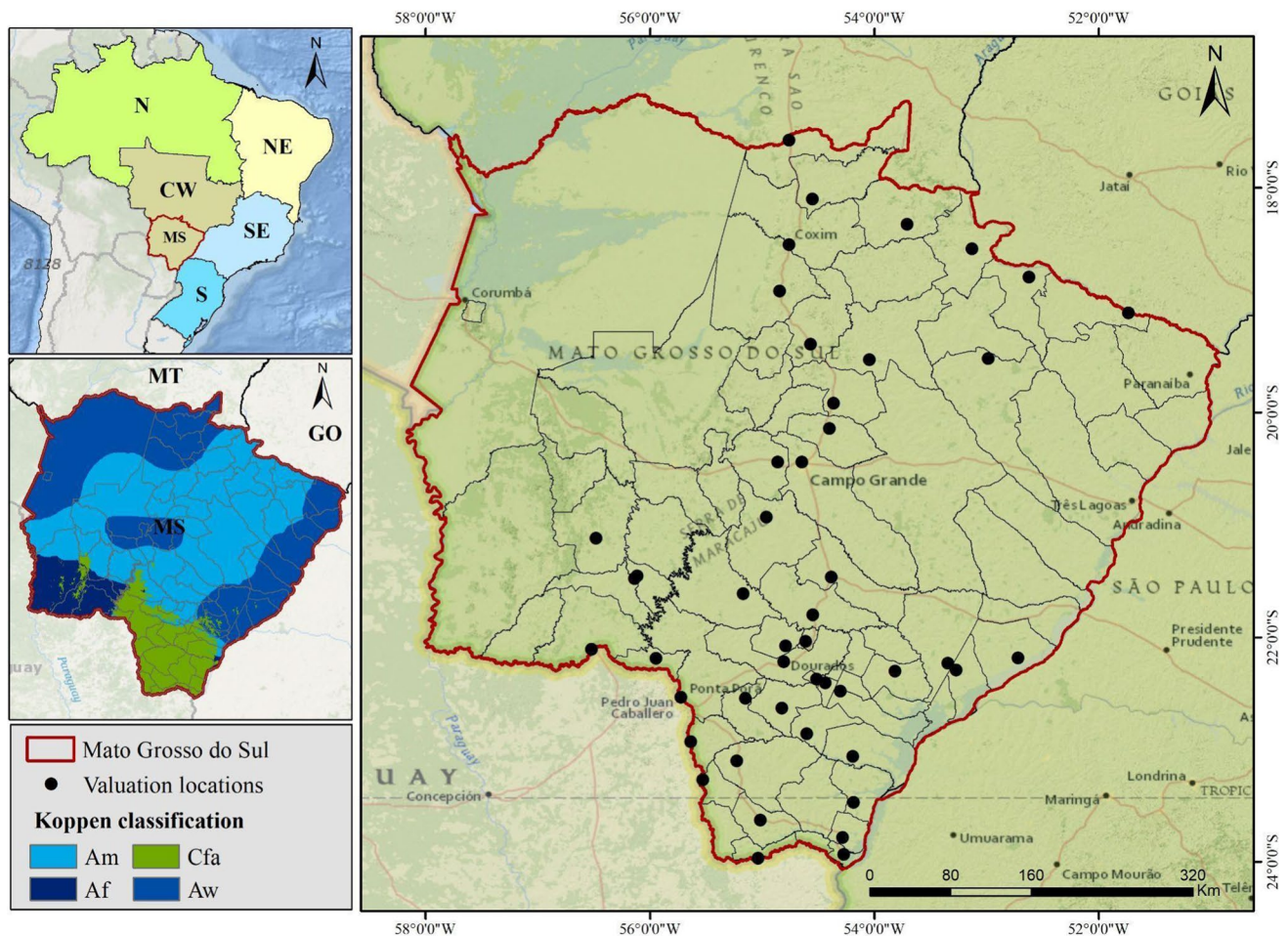
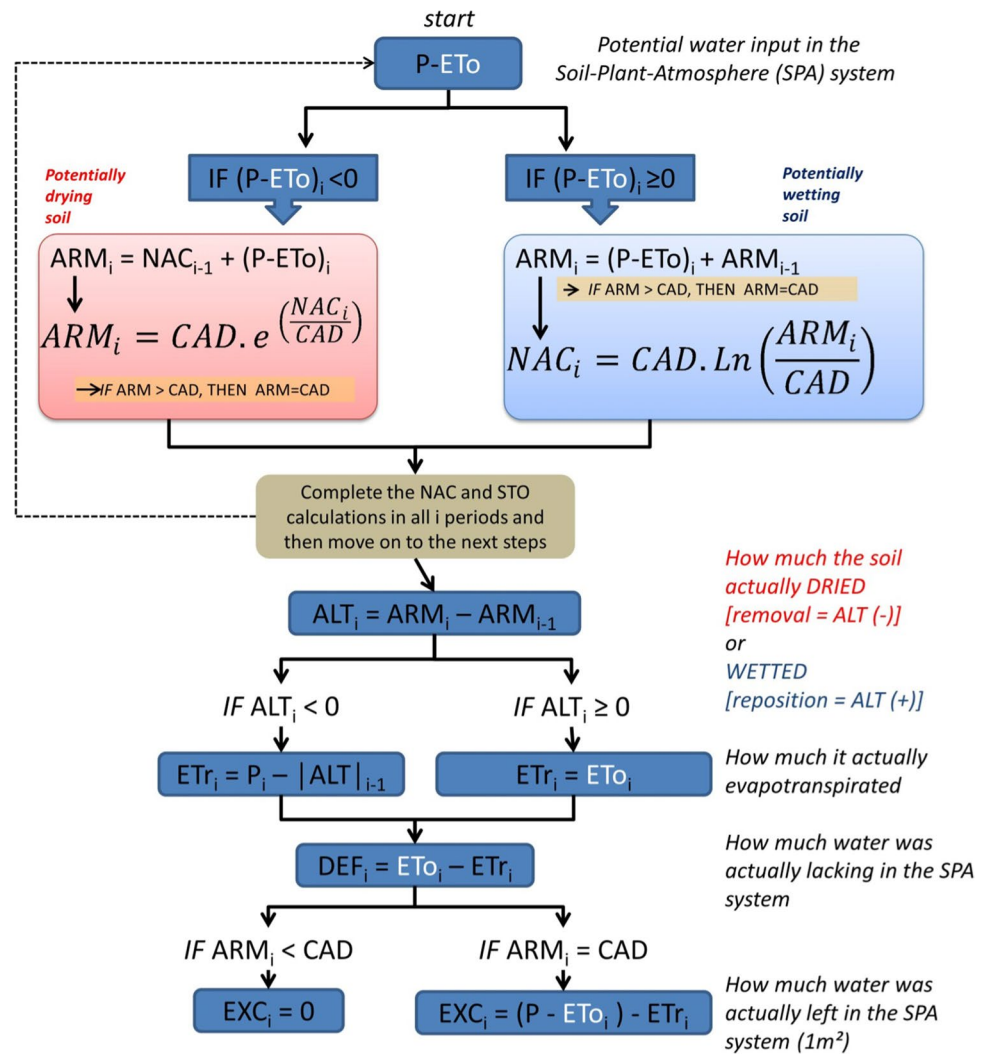


Fig. 1 Geographic location of soybean-producing regions in Mato Grosso do Sul, Brazil



**Fig. 2** Flowchart of the water balance model modified from Thornthwaite and Mather (1955) where in CAD is the available water capacity (mm), ARM is the soil water storage (mm), EXC is the water surplus in the soil–plant–atmosphere system (mm), DEF is the water deficit of the soil–plant–atmosphere system (mm), NAC is the negative P-ET<sub>o</sub> value, P is the precipitation (mm), ET<sub>o</sub> is the potential evapotranspiration (mm), ETR is the real evapotranspiration (mm), ALT is the alteration in soil water storage for the current and previous months (mm), and *i* is the current month. Adapted by Aparecido et al. (2020)



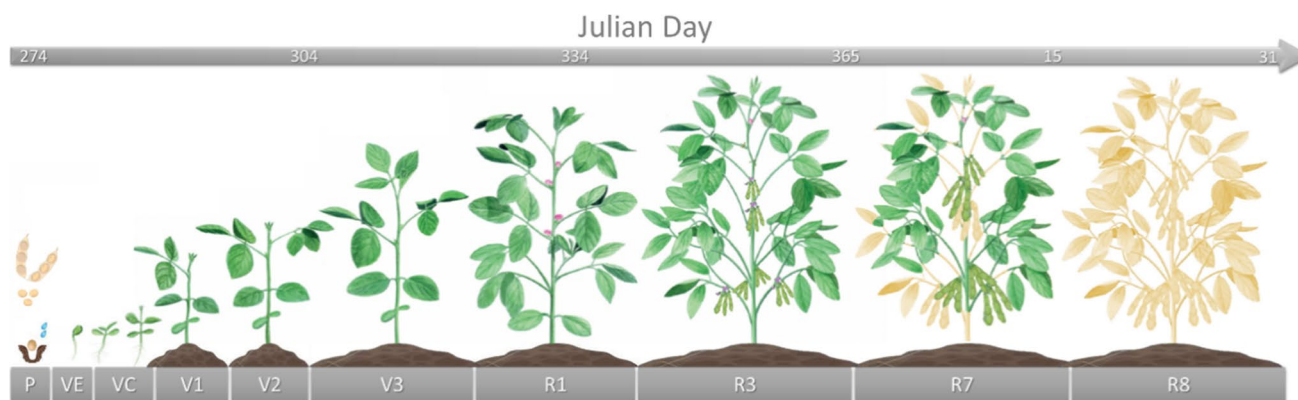
**Table 1** Parameters defined and tested in each machine learning algorithm used to predict soybean yields in Mato Grosso do Sul

| Model        | Parameter          | Values tested                                                                       | Defined values |
|--------------|--------------------|-------------------------------------------------------------------------------------|----------------|
| MLR          | k_features         | 1; 15; 30; 45; 70                                                                   | 70             |
|              | cv                 | 2; 3; 4; 5                                                                          | 3              |
| MLP          | Hidden_layer_sizes | (9, 10, 9); (7, 8, 10); (10, 10, 10); (25, 11, 7, 5, 3); (10, 10, 10); (20, 20, 20) | 10, 10, 10     |
|              | Max_iter           | 100; 1000; 2000; 3000; 5000; 7000                                                   | 1000           |
|              | cv                 | 2; 3; 4; 10; 15; 20                                                                 | 2              |
| SVM          | Kernel             | rbf; linear; poly                                                                   | rbf            |
|              | c                  | 50; 100; 1000; 10,000                                                               | 100            |
|              | Epsilon            | 1; 2; 3; 4; 5; 10                                                                   | 4              |
| RF           | Max_depth          | none; 15; 30; 300                                                                   | none           |
|              | n_estimators       | 10; 200; 500; 1000; 10,000                                                          | 500            |
|              | Random_state       | 2; 10; 20; 50                                                                       | 10             |
| XGBOOSTING   | Max_depth          | 1; 10; 100                                                                          | 10             |
|              | Alpha              | 20; 200; 2000                                                                       | 20             |
| GradBOOSTING | n_estimators       | 300; 3000; 30,000                                                                   | 300            |
|              | n_estimators       | 10; 1000; 3000; 10,000                                                              | 1000           |

**Table 2** Daily Pearson correlations between climatic variables and soybean yield in Mato Grosso do Sul in October, November, and December. Legend: Blue is positive correlation and Red negative correlation; ARM, soil water storage; DEF, water deficit; DPM, leaf wetness duration; ETP, potential evapotranspiration; ETR, actual evapotranspiration; EXC, water surplus; P, precipitation; Qg, incident solar radiation; rhmax, maximum relative humidity of the day; rhmin, minimum relative humidity of the day; tdew, dew point temperature; TMAX, maximum temperature; TMED, average temperature; TMIN, minimum temperature; UR, relative humidity of the air; VMAX, maximum wind speed; VMED, average wind speed; VMIN, minimum wind speed

| DAYS     | Climate Variables |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
|----------|-------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|          | ARM               | DEF   | DPM   | ETP   | ETR   | EXC   | P     | QG    | RHMAX | RHMIN | TDEW  | TMAX  | TMED  | TMIN  | UR    | VMAX  | VMED  | VMIN  |
| OCTOBER  |                   |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| 1        | -0.32             | 0.18  | -0.18 | 0.01  | -0.16 | -0.05 | 0.06  | 0     | -0.22 | -0.23 | -0.25 | 0.15  | 0.12  | 0.01  | -0.24 | 0.33  | 0.34  | 0.26  |
| 2        | -0.22             | 0.15  | -0.1  | -0.12 | -0.27 | -0.03 | 0.26  | -0.12 | -0.14 | -0.1  | -0.41 | -0.09 | -0.12 | -0.18 | -0.16 | 0.26  | 0.23  | 0.06  |
| 3        | -0.25             | 0.09  | -0.24 | -0.04 | -0.15 | -0.03 | -0.12 | -0.03 | -0.27 | -0.21 | -0.33 | -0.02 | -0.1  | -0.17 | -0.25 | 0.06  | 0.04  | 0.01  |
| 4        | -0.27             | 0.14  | -0.23 | 0.05  | -0.1  | -0.1  | -0.11 | 0.07  | -0.2  | -0.18 | -0.2  | 0.06  | -0.01 | -0.07 | -0.2  | -0.06 | -0.04 | -0.01 |
| 5        | -0.24             | 0.04  | 0     | -0.05 | -0.08 | -0.02 | 0.24  | -0.07 | -0.03 | -0.03 | 0.06  | 0.14  | 0.13  | 0.07  | -0.07 | 0.15  | 0.13  | 0.05  |
| 6        | -0.24             | 0.12  | -0.19 | -0.01 | -0.14 | -0.13 | 0.06  | -0.02 | -0.16 | -0.15 | -0.05 | 0.09  | 0.08  | 0.06  | -0.17 | 0.25  | 0.26  | 0.09  |
| 7        | -0.24             | 0.06  | -0.14 | 0     | -0.07 | -0.13 | -0.1  | -0.03 | -0.1  | -0.17 | -0.09 | 0.08  | 0.04  | -0.01 | -0.15 | -0.02 | -0.05 | -0.13 |
| 8        | -0.25             | 0.13  | -0.16 | 0.11  | -0.03 | -0.25 | -0.24 | 0.12  | -0.13 | -0.11 | -0.12 | 0     | -0.03 | -0.03 | -0.11 | -0.07 | -0.01 | 0.04  |
| 9        | -0.31             | 0.28  | -0.2  | 0.14  | -0.2  | -0.28 | -0.31 | 0.14  | -0.17 | -0.2  | -0.2  | 0.07  | 0.02  | -0.06 | -0.21 | -0.05 | 0.01  | 0.05  |
| 10       | -0.33             | 0.3   | -0.37 | 0.3   | 0.09  | -0.23 | -0.28 | 0.31  | -0.27 | -0.29 | -0.21 | 0.21  | 0.17  | 0.02  | -0.3  | 0.23  | 0.24  | 0.2   |
| 11       | -0.32             | 0.19  | -0.3  | 0.16  | -0.1  | -0.02 | -0.02 | 0.15  | -0.23 | -0.24 | -0.09 | 0.2   | 0.18  | 0.13  | -0.25 | 0.21  | 0.16  | 0.16  |
| 12       | -0.33             | 0.22  | -0.37 | 0.12  | -0.01 | -0.04 | -0.13 | 0.1   | -0.29 | -0.23 | -0.13 | 0.15  | 0.16  | 0.15  | -0.26 | 0.21  | 0.15  | 0.06  |
| 13       | -0.33             | 0.11  | -0.3  | 0.02  | -0.07 | -0.1  | -0.05 | 0.02  | -0.18 | -0.26 | -0.15 | 0.2   | 0.14  | 0.05  | -0.23 | 0.19  | 0.21  | 0.16  |
| 14       | -0.29             | 0.07  | -0.31 | -0.07 | -0.22 | -0.15 | 0.07  | -0.05 | -0.17 | -0.18 | -0.15 | 0.11  | 0.06  | 0.03  | -0.17 | 0.18  | 0.25  | 0.26  |
| 15       | -0.31             | 0.23  | -0.23 | 0.2   | -0.05 | -0.05 | -0.15 | 0.24  | -0.12 | -0.24 | -0.11 | 0.2   | 0.17  | 0.02  | -0.22 | 0.08  | 0.08  | 0.17  |
| 16       | -0.31             | 0.2   | -0.32 | 0.08  | -0.03 | -0.1  | -0.09 | 0.09  | -0.25 | -0.24 | -0.11 | 0.25  | 0.25  | 0.24  | -0.24 | 0.33  | 0.31  | 0.23  |
| 17       | -0.31             | 0.26  | -0.3  | 0.08  | -0.22 | -0.12 | 0     | 0.01  | -0.24 | -0.24 | -0.09 | 0.24  | 0.25  | 0.2   | -0.25 | 0.22  | 0.21  | 0.12  |
| 18       | -0.32             | 0.23  | -0.27 | 0.14  | -0.18 | -0.18 | -0.19 | 0     | -0.24 | -0.2  | 0.04  | 0.22  | 0.23  | 0.24  | -0.22 | 0.1   | 0.07  | 0.06  |
| 19       | -0.33             | 0.1   | -0.33 | -0.01 | -0.13 | -0.09 | -0.05 | -0.12 | -0.29 | -0.26 | 0.04  | 0.33  | 0.31  | 0.23  | -0.28 | 0.19  | 0.13  | 0.02  |
| 20       | -0.31             | 0.02  | -0.19 | -0.14 | -0.17 | -0.01 | 0.15  | -0.21 | -0.17 | 0.04  | 0.15  | 0.1   | 0.21  | 0.25  | -0.04 | -0.02 | -0.06 | -0.18 |
| 21       | -0.29             | -0.01 | -0.18 | -0.11 | -0.14 | -0.06 | 0.07  | -0.19 | -0.16 | 0.03  | 0.13  | 0.09  | 0.17  | 0.25  | -0.04 | -0.03 | 0.02  | -0.02 |
| 22       | -0.16             | -0.06 | -0.01 | -0.26 | -0.25 | -0.02 | 0.2   | -0.32 | -0.05 | 0.21  | 0.07  | -0.17 | -0.04 | 0.11  | 0.11  | -0.02 | 0.03  | -0.04 |
| 23       | -0.19             | 0.05  | -0.14 | 0.02  | 0.01  | -0.06 | -0.08 | 0.07  | -0.13 | -0.04 | -0.04 | 0     | 0.02  | 0.05  | -0.06 | -0.02 | -0.04 | -0.06 |
| 24       | -0.25             | 0.29  | -0.18 | 0.12  | -0.14 | -0.25 | -0.35 | 0.1   | -0.15 | -0.18 | -0.03 | 0.21  | 0.21  | 0.09  | -0.18 | 0.09  | 0.09  | 0.09  |
| 25       | -0.26             | 0.22  | -0.29 | 0.28  | 0.08  | -0.26 | -0.19 | 0.29  | -0.3  | -0.18 | -0.03 | 0.19  | 0.27  | 0.29  | -0.22 | 0.08  | 0.1   | 0.01  |
| 26       | -0.15             | -0.11 | -0.17 | -0.11 | -0.05 | -0.11 | 0.02  | 0.1   | -0.15 | -0.14 | 0     | 0.16  | 0.17  | 0.13  | -0.16 | 0.13  | 0.12  | 0.02  |
| 27       | -0.12             | -0.03 | -0.25 | -0.09 | -0.07 | 0     | 0.14  | -0.08 | -0.28 | -0.16 | -0.1  | 0.09  | 0.06  | 0.07  | -0.18 | 0.29  | 0.27  | 0.14  |
| 28       | -0.1              | -0.03 | -0.12 | -0.04 | -0.06 | -0.02 | 0.18  | -0.06 | -0.19 | 0.02  | -0.09 | -0.07 | -0.06 | 0.03  | -0.02 | 0.33  | 0.31  | 0.13  |
| 29       | -0.09             | 0.04  | -0.06 | -0.09 | -0.13 | 0.01  | -0.02 | -0.04 | -0.12 | -0.1  | -0.19 | -0.04 | -0.05 | -0.07 | -0.13 | 0.11  | 0.13  | 0.05  |
| 30       | -0.08             | 0.06  | -0.19 | -0.05 | -0.18 | -0.04 | -0.02 | -0.05 | -0.15 | -0.16 | -0.14 | 0.11  | 0.06  | 0.02  | -0.14 | 0.22  | 0.17  | -0.07 |
| 31       | -0.1              | 0.01  | -0.06 | -0.09 | -0.14 | -0.18 | -0.17 | -0.12 | -0.07 | -0.03 | -0.03 | 0.01  | 0.03  | 0.03  | -0.05 | 0.16  | 0.16  | 0.02  |
| NOVEMBER |                   |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| 1        | -0.13             | 0.2   | -0.15 | 0.28  | 0.19  | -0.25 | -0.26 | 0.26  | -0.12 | -0.27 | -0.09 | 0.21  | 0.11  | -0.01 | -0.25 | 0.21  | 0.22  | 0.21  |
| 2        | -0.11             | 0.09  | -0.11 | -0.09 | -0.22 | -0.01 | 0.02  | -0.1  | -0.09 | -0.19 | -0.06 | 0.16  | 0.1   | 0     | -0.19 | 0.23  | 0.24  | 0.19  |
| 3        | -0.12             | 0.09  | -0.24 | -0.01 | -0.12 | -0.06 | -0.11 | 0     | -0.21 | -0.14 | -0.08 | 0.07  | 0.04  | 0.04  | -0.15 | 0.18  | 0.22  | 0.15  |
| 4        | -0.08             | -0.04 | -0.04 | -0.1  | -0.08 | 0.05  | 0.18  | -0.1  | 0     | -0.02 | -0.02 | 0.01  | 0.02  | -0.01 | -0.03 | 0.24  | 0.24  | 0.11  |
| 5        | -0.05             | 0.02  | -0.02 | -0.12 | -0.18 | 0.07  | 0.13  | -0.14 | -0.06 | 0.02  | -0.01 | 0.01  | 0.02  | 0.04  | -0.01 | 0.13  | 0.15  | 0.13  |
| 6        | -0.05             | 0.01  | -0.11 | 0.03  | 0.02  | -0.02 | -0.01 | 0.01  | -0.15 | -0.04 | 0.01  | 0.06  | 0.06  | 0.09  | -0.07 | 0.1   | 0.03  | -0.08 |
| 7        | -0.04             | 0.08  | -0.1  | 0.08  | 0.01  | 0.1   | 0.08  | 0.05  | -0.1  | -0.08 | -0.04 | 0.09  | 0.08  | 0.04  | -0.11 | 0.07  | 0.04  | -0.07 |
| 8        | -0.02             | -0.04 | -0.05 | -0.2  | -0.23 | 0.16  | 0.13  | -0.26 | -0.08 | -0.06 | 0.1   | 0.16  | 0.2   | 0.13  | -0.08 | -0.02 | -0.09 | -0.12 |
| 9        | 0.01              | -0.03 | -0.08 | -0.02 | 0.02  | 0.07  | 0.18  | 0.02  | -0.16 | 0.07  | 0.16  | 0.01  | 0.13  | 0.25  | -0.02 | -0.05 | -0.06 | -0.16 |
| 10       | 0.04              | -0.11 | 0.01  | -0.02 | 0.09  | 0.14  | 0.2   | -0.03 | -0.06 | 0.17  | 0.12  | -0.07 | 0.02  | 0.15  | 0.1   | -0.01 | 0.03  | -0.03 |
| 11       | 0.03              | -0.1  | -0.19 | -0.03 | 0.02  | -0.04 | -0.04 | 0.02  | -0.2  | -0.01 | -0.19 | -0.16 | -0.13 | -0.01 | -0.07 | 0.04  | 0.08  | 0.02  |
| 12       | 0.02              | -0.08 | -0.14 | -0.05 | 0.01  | -0.06 | -0.06 | 0.02  | -0.11 | -0.08 | -0.26 | -0.17 | -0.2  | -0.2  | -0.1  | 0.06  | 0.03  | 0.01  |
| 13       | -0.02             | -0.09 | -0.12 | -0.01 | 0.07  | -0.19 | -0.15 | 0.02  | -0.11 | -0.14 | -0.25 | -0.07 | -0.17 | -0.2  | -0.14 | -0.02 | -0.01 | -0.05 |
| 14       | -0.08             | 0.04  | -0.2  | 0.19  | 0.24  | -0.19 | -0.22 | 0.19  | -0.18 | -0.25 | -0.21 | 0.18  | 0.06  | -0.13 | -0.25 | 0.03  | -0.03 | -0.02 |
| 15       | -0.07             | 0.09  | -0.19 | 0     | -0.08 | -0.01 | 0     | -0.03 | -0.17 | -0.15 | -0.03 | 0.17  | 0.18  | 0.08  | -0.19 | 0.07  | 0.02  | -0.01 |
| 16       | -0.08             | 0.03  | -0.2  | 0.12  | 0.13  | -0.04 | -0.03 | 0.11  | -0.17 | -0.11 | 0.02  | 0.16  | 0.19  | 0.14  | -0.13 | 0.34  | 0.28  | 0.1   |
| 17       | -0.05             | -0.07 | -0.19 | -0.06 | 0.04  | -0.02 | 0.09  | -0.06 | -0.14 | -0.05 | 0.02  | 0.09  | 0.12  | 0.13  | -0.06 | 0.21  | 0.23  | 0.12  |
| 18       | 0                 | -0.06 | -0.07 | -0.09 | -0.07 | -0.04 | 0.09  | -0.07 | -0.07 | 0.01  | -0.05 | -0.09 | -0.07 | -0.02 | -0.01 | 0.19  | 0.25  | 0.19  |
| 19       | -0.03             | 0.08  | -0.09 | 0     | -0.1  | 0.01  | -0.1  | 0.02  | -0.07 | -0.09 | -0.12 | 0.02  | -0.03 | -0.09 | -0.11 | 0.02  | 0.04  | 0.05  |
| 20       | -0.02             | 0.1   | -0.12 | 0.08  | -0.02 | 0.02  | 0.06  | -0.11 | -0.17 | -0.12 | 0.16  | 0.13  | -0.04 | -0.19 | 0.01  | -0.01 | -0.09 | -0.09 |
| 21       | -0.05             | -0.01 | -0.14 | -0.03 | -0.03 | 0.01  | -0.09 | -0.07 | -0.14 | -0.06 | -0.06 | 0.05  | 0.08  | 0.06  | -0.09 | -0.01 | 0.03  | 0     |
| 22       | -0.06             | 0.01  | -0.16 | -0.01 | -0.02 | -0.01 | -0.03 | 0.01  | -0.13 | -0.14 | -0.14 | 0.08  | 0.05  | 0     | -0.14 | 0.08  | 0.06  | 0.04  |
| 23       | -0.12             | 0.15  | -0.17 | 0.26  | 0.14  | -0.04 | -0.19 | 0.26  | -0.09 | -0.32 | -0.23 | 0.26  | 0.12  | -0.3  | -0.29 | -0.01 | -0.03 | -0.05 |
| 24       | -0.12             | 0.13  | -0.18 | 0.12  | -0.05 | 0     | -0.01 | 0.04  | -0.1  | -0.18 | -0.05 | 0.24  | 0.24  | 0.05  | -0.2  | 0.02  | 0     | -0.02 |
| 25       | -0.08             | -0.08 | -0.15 | -0.16 | -0.12 | 0.03  | 0.12  | -0.19 | -0.13 | -0.02 | 0.12  | 0.13  | 0.21  | 0.21  | -0.06 | 0.24  | 0.2   | 0.07  |
| 26       | -0.07             | -0.08 | -0.09 | -0.24 | -0.24 | -0.02 | 0.02  | -0.25 | -0.16 | 0.04  | 0.03  | -0.03 | 0.03  | 0.16  | 0.01  | 0.14  | 0.15  | 0.04  |
| 27       | -0.04             | -0.2  | -0.08 | -0.21 | -0.11 | 0.03  | 0.1   | -0.21 | -0.07 | 0.03  | 0.06  | -0.02 | 0     | 0.13  | 0.04  | 0.15  | 0.13  | 0.05  |
| 28       | 0.01              | -0.14 | 0.01  | -0.12 | 0.03  | -0.07 | 0.04  | 0.1   | 0     | 0.07  | 0.06  | -0.09 | -0.07 | 0.06  | 0.07  | 0.1   | 0.1   | -0.04 |
| 29       | -0.01             | -0.07 | 0.02  | 0.05  | 0.12  | 0.04  | 0     | 0.06  | -0.01 | 0.01  | 0.04  | 0     | 0.03  | 0.06  | 0     | 0.13  | 0.07  | -0.04 |
| 30       | 0.05              | -0.19 | 0.1   | -0.16 | -0.05 | 0.28  | 0.35  | -0.14 | 0.06  | 0.15  | 0.07  | -0.15 | -0.12 | 0.02  | 0.13  | 0.18  | 0.16  | -0.02 |
| DECEMBER |                   |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| 1        | 0.05              | -0.02 | 0.02  | -0.16 | -0.14 | 0.07  | 0.02  | -0.15 | 0.02  | 0.05  | 0.02  | -0.06 | -0.07 | 0.01  | 0.08  | 0.07  | 0.04  | -0.06 |
| 2        | 0.06              | -0.17 | 0.05  | -0.11 | 0.07  | -0.01 | 0.06  | -0.13 | 0.14  | 0.16  | 0.21  | -0.03 | 0.11  | 0.19  | 0.16  | 0.17  | 0.07  | -0.05 |
| 3        | 0.08              | -0.23 | 0.05  | -0.02 | 0.07  | -0.03 | 0.14  | -0.03 | 0.03  | 0.24  | 0.22  | -0.14 | 0.07  | 0.24  | 0.19  | 0.19  | 0.12  | -0.02 |
| 4        | 0.11              | -0.2  | 0.07  | -0.26 | -0.09 | -0.05 | 0.01  | -0.26 | 0     | 0.27  | 0.24  | -0.19 | 0.01  | 0.26  | 0.22  | 0.04  | 0.03  | -0.06 |
| 5        | 0.14              | -0.08 | 0.13  | -0.07 | -0.03 | 0.07  | 0.13  | -0.09 | 0.1   | 0.15  | 0.21  | -0.08 | 0.01  | 0.1   | 0.15  | 0     | 0     | -0.03 |
| 6        | 0.16              | -0.1  | 0.13  | -0.17 | -0.11 | -0.27 | 0.19  | -0.19 | 0.09  | 0.2   | 0.26  | -0.06 | 0.08  | 0.2   | 0.1   |       |       |       |





**Fig. 3** Average phenology for soybean cultivated in Mato Grosso do Sul. Legend: P is planting, VE is emergence, VC is the cotyledon stage, V1 is one vegetative node formed, V2 is two vegetative nodes

formed, V3 is three vegetative nodes formed, R1 is the beginning of flowering, R3 is the beginning of pod formation, R7 is beginning of maturation, and R8 is full maturation

1979). Table 2 shows the tested and defined parameters of each algorithm.

### 2.3 2nd stage—application of calibrated MLs throughout the MS state

The topologies of the selected MLs with the best performances in the first stage of the project were pre-selected to predict the soybean yield of the entire MS (2nd stage). The most accurate MLs for the entire MS were applied in this 2nd stage, using data from all municipalities together. The same procedure for collecting meteorological data by the NASA-POWER system and calculating ETo and WB

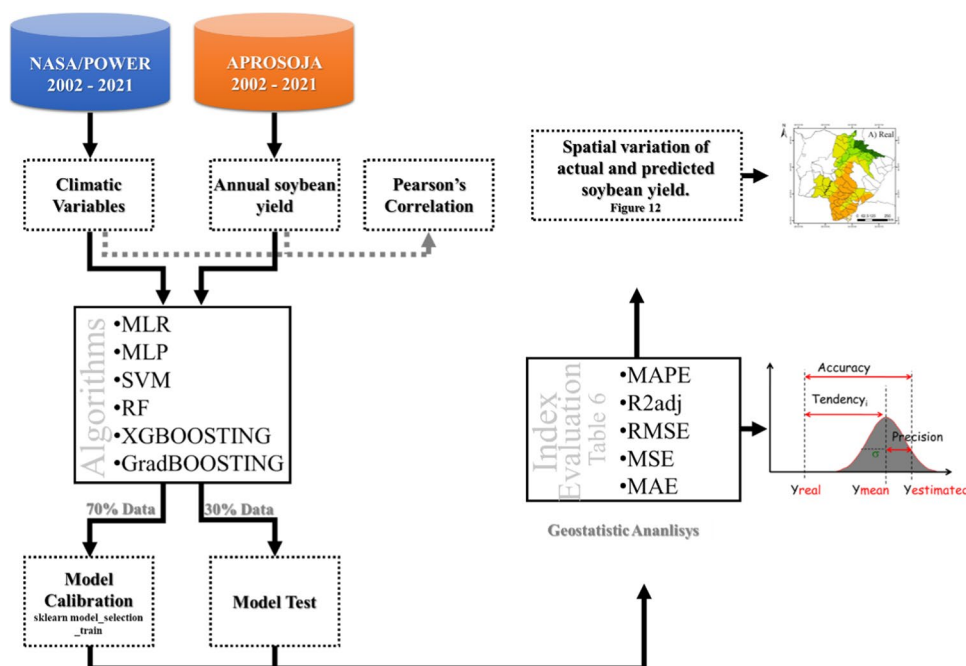
was performed at this stage of the study, but the data were obtained for all locations in the MS state, referring to the period from 1981 to 2021 (Fig. 1).

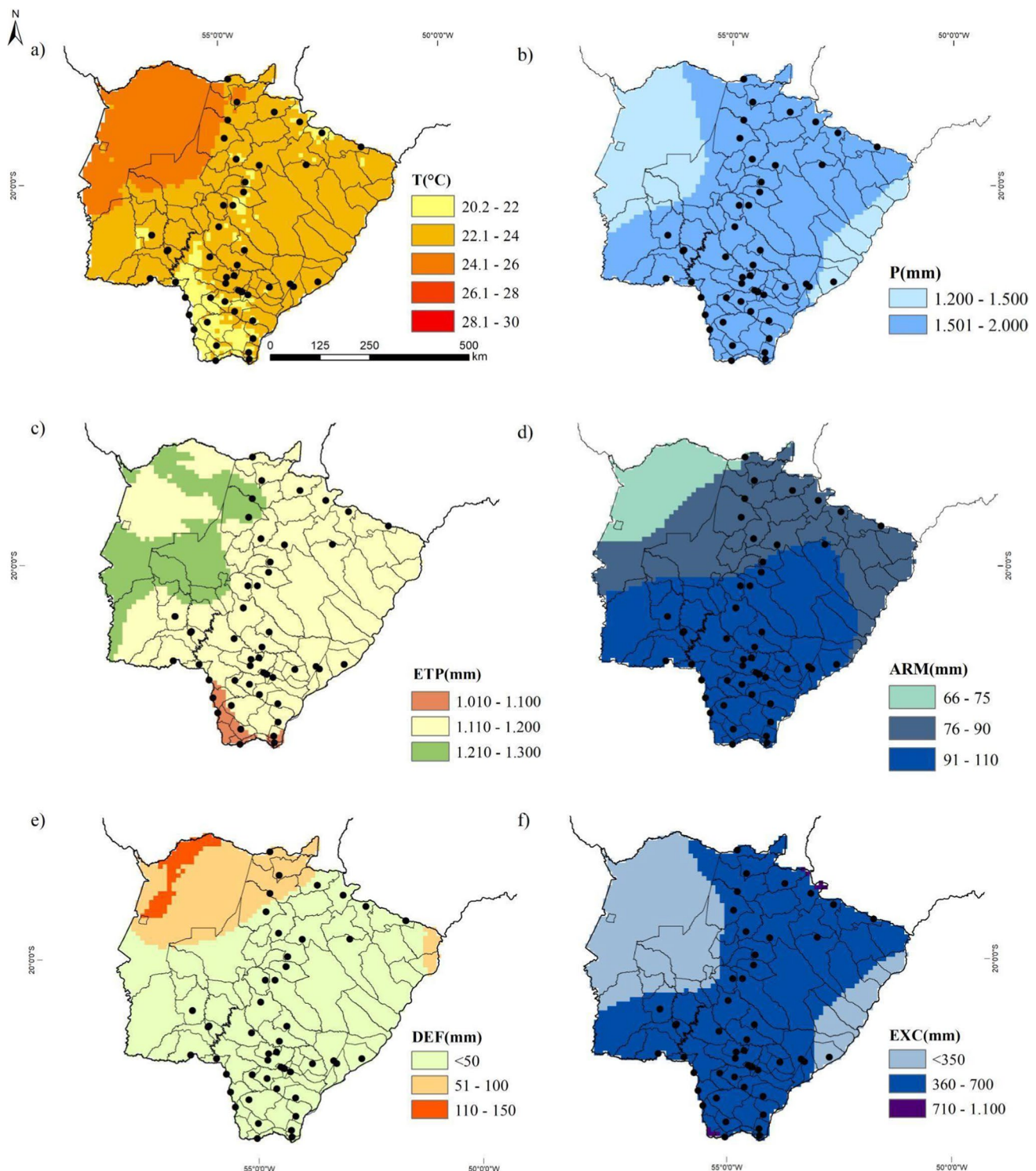
This process was conducted in Python programming language (<https://www.python.org/>, version 3.7.3 or higher), allowing work with large datasets.

Definition of input variables (FEATURES) in machine learning models.

The FEATURES correspond to the independent variables of the algorithms, while soybean yield was the dependent variable. The soybean planting period was considered to define the input climate variables (Fig. 3). The FEATURES were organized as follows: T-2 corresponding to the air temperature

**Fig. 4** Detailed flowchart of all stages of this work

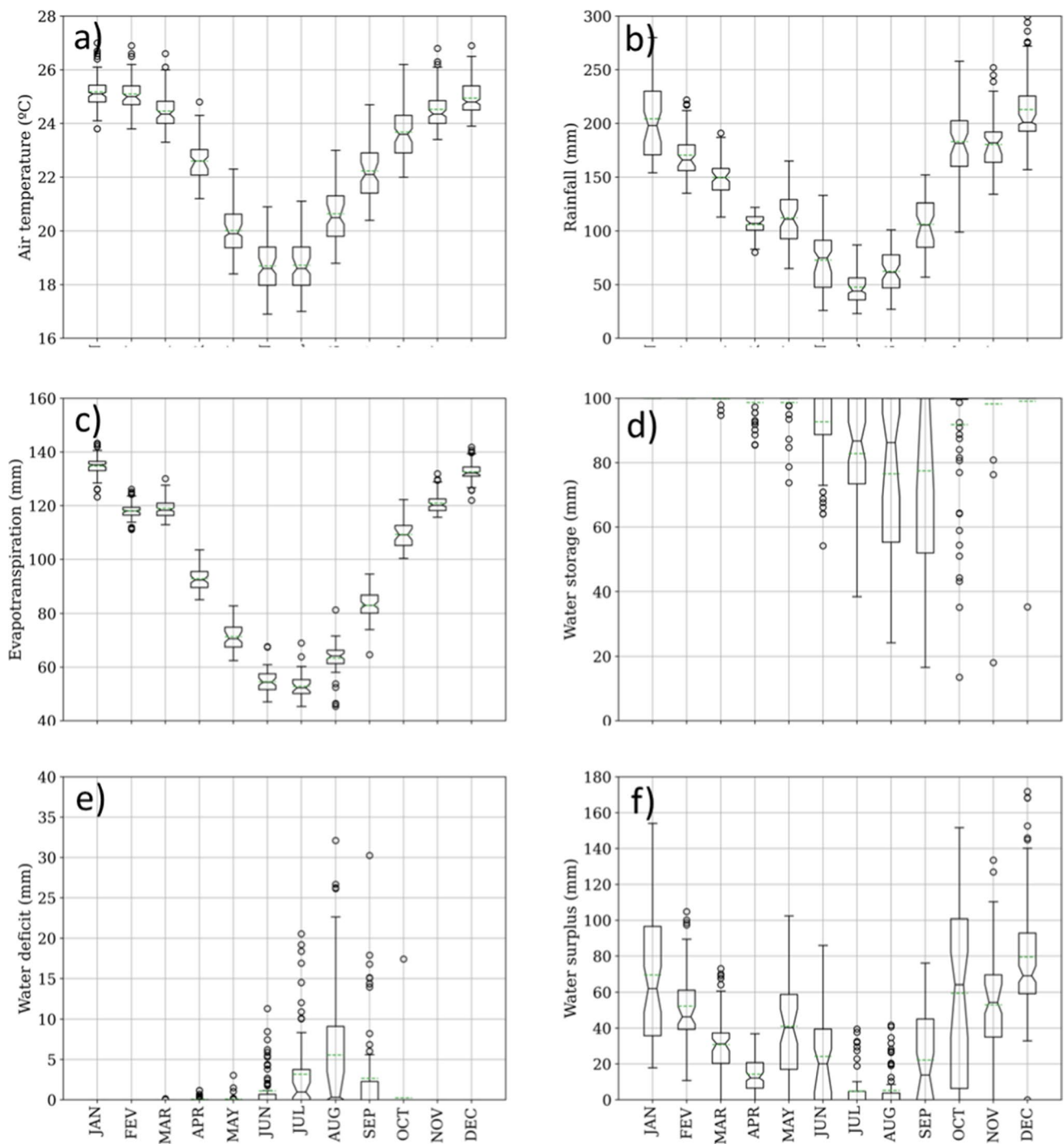




**Fig. 5** Spatial variability of climatic elements **A** air temperature, **B** precipitation, **C** evapotranspiration, **D** soil water storage, **E** water deficiency, and **F** water surplus, for the state of Mato Grosso do Sul, Brazil

on the 2nd day before the soybean harvest period, similarly for T-9, T-8... T0. The same procedure was performed for P, Qg, u2, Rn, RH, ETo, ETr, ARM, DEF, and EXC, initially corresponding to the input layer of neural networks.

The crop develops best at temperatures ranging from 16 to 27 °C (Major et al. 1975; van Schaik and Probst 1958). However, the ideal temperature for its development during germination and emergence is around 30 °C (Gibson and



**Fig. 6** Seasonal variability of climatic elements **A** air temperature, **B** precipitation, **C** evapotranspiration, **D** soil water storage, **E** water deficiency, and **F** water surplus, for the state of Mato Grosso do Sul, Brazil. Average data from 47 locations

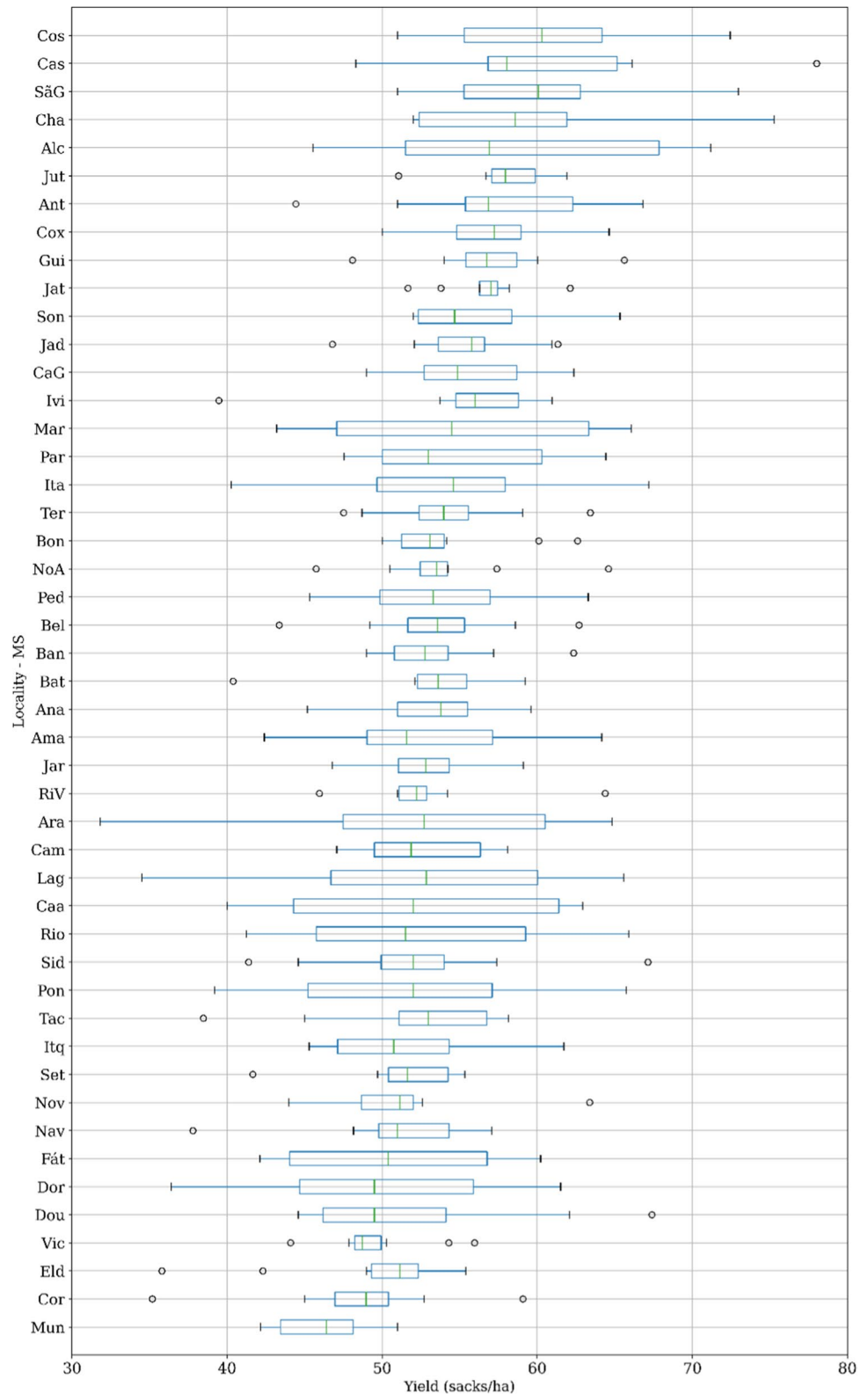
Mullen 1996). Regions with temperatures lower than 10 °C are not suitable for cultivation because the vegetative growth is small or null (Hopper et al. 1979). However, growth rate,

flowering, and pod retention are impaired at temperatures higher than 40 °C (Gibson and Mullen 1996).

Water makes up approximately 90% of the soybean plant mass, playing roles in virtually all physiological and



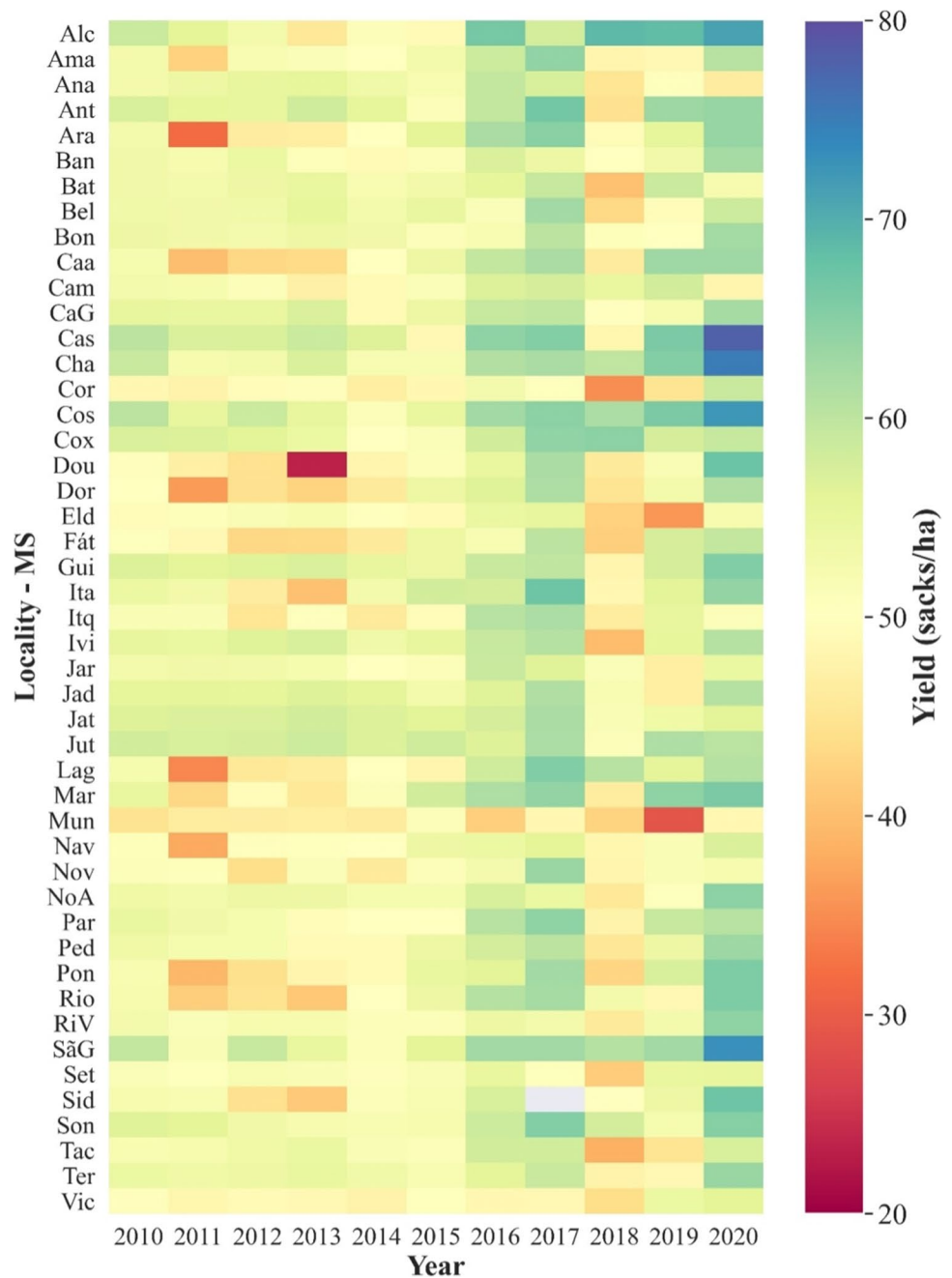
**Fig. 7** Boxplot of soybean yield variation in the main producing cities in Mato Grosso do Sul from 2010 to 2021. Legend: Vic, Vicentina; Ter, Terenos; Tac, Tacuru; Son, Sonora; Sid, Sidrolândia; Set, Sete Quedas; SãG, São Gabriel do Oeste; RiV, Rio Verde de MT; Rio, Rio Brilhante; Pon, Ponta Porã; Ped, Pedro Gomes; Par, Paraíso das Águas; NoA, Nova Andradina; Nov, Nova Alvorada do Sul; Nav, Naviraí; Mun, Mundo Novo; Mar, Maracaju; Lag, Laguna Carapã; Jut, Juti; Jat, Jateí; Jad, Jardim; Jar, Jaraguari; Ivi, Ivinhema; Itq, Itaquiraí; Ita, Itaporã; Gui, Guia Lopes da Laguna; Fát, Fátima do Sul; Eld, Eldorado; Dor, Dourados; Dou, Douradina; Cox, Coxim; Cos, Costa Rica; Cor, Coronel Sapucaia; Cha, Chapadão do Sul; Cas, Cassilândia; CaG, Campo Grande; Cam, Camapuã; Caa, Caarapó; Bon, Bonito; Bel, Bela vista; Bat, Batayporã; Ban, Bandeirantes; Ara, Aral Moreira; Ant, Antônio João; Ana, Anaurilândia; Ama, Amambai; Alc, Alcinoópolis



biochemical processes (García-Villalba et al. 2008). During the first period, both the excess and the deficit of water are harmful to the crop establishment and the achievement of a good uniformity in the plant population, and the excess water is

more limiting than the deficit. Soybean seeds need to absorb at least 50% of their weight in water to ensure good germination. The water content in the soil at this stage must not exceed 85% of the maximum total available, nor be less than 50%. Water

**Fig. 8** Heat map of the temporal variation of soybean yield in the main producing cities in Mato Grosso do Sul from 2010/2011 to 2020/2021. Legend: Vic, Vicentina; Ter, Terenos; Tac, Tacuru; Son, Sonora; Sid, Sidrolândia; Set, Sete Quedas; SãG, São Gabriel do Oeste; RiV, Rio Verde de MT; Rio, Rio Brilhante; Pon, Ponta Porã; Ped, Pedro Gomes; Par, Paraíso das Águas; NoA, Nova Andradina; Nov, Nova Alvorada do Sul; Nav, Naviraí; Mun, Mundo Novo; Mar, Maracaju; Lag, Laguna Carapã; Jut, Juti; Jat, Jateí; Jad, Jardim; Jar, Jaraguari; Ivi, Ivinhema; Itq, Itaquiraí; Ita, Itaporã; Gui, Guia Lopes da Laguna; Fát, Fátima do Sul; Eld, Eldorado; Dor, Dourados; Dou, Douradina; Cox, Coxim; Cos, Costa Rica; Cor, Coronel Sapucaia; Cha, Chapadão do Sul; Cas, Cassilândia; CaG, Campo Grande; Cam, Camapuã; Caa, Caarapó; Bon, Bonito; Bel, Bela vista; Bat, Batayporã; Ban, Bandeirantes; Ara, Aral Moreira; Ant, Antônio João; Ana, Anaurilândia; Ama, Amambai; Alc, Alcinoópolis

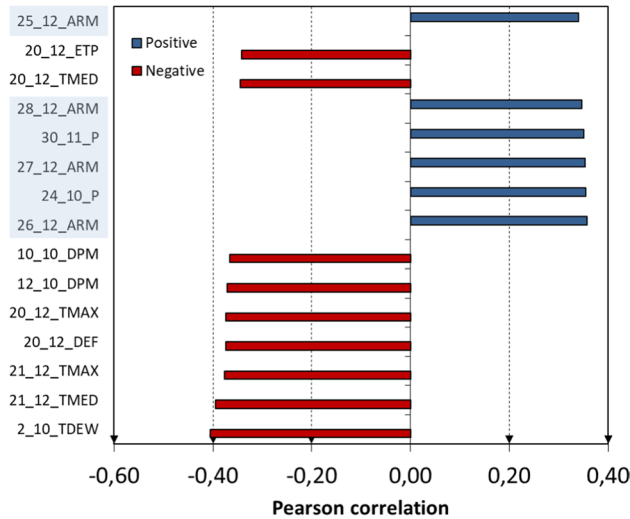


requirement in soybean crops increases with plant development, reaching a maximum during flowering-grain filling (7 to 8 mm/day). Overall, the highest water consumption coincides with the period in which the crop has the highest height and leaf area index. The total water requirement of soybean crops to obtain maximum yield varies from 450 to 800 mm/cycle, depending on climate conditions, crop management, and cycle duration (Tacker and Vories 2014). Water consumption by the soybean crop depends not only on the developmental stage but also on the evaporative demand of the atmosphere, and

its absolute value may vary both as a function of the climate conditions of each region and as a function of the year and sowing season (weather conditions) in the same climate region.

## 2.4 Model performance evaluation

The metrics used to verify the degree of learning consisted of the mean square error (MSE), the performance evaluation of MLs in the validation period used the accuracy through the Mean Absolute Percent Error (MAPE) and mean



**Fig. 9** Values of the 15 greatest correlations between climate variables and soybean yield in Mato Grosso do Sul. Caption: 25\_12\_ARM is Soil water storage on December 25th, 21\_12\_TMED December 21st average temperature, 21\_12\_TMAX December 21st maximum temperature, 20\_12\_DEF December 20th water deficit, 20\_12\_TMAX December 20th maximum temperature, 12\_10\_DPM October 12th leaf wetness duration, 10\_10\_DPM October 10th leaf wetness duration, 26\_12\_ARM is Dec 26 Soil water storage, 24\_10\_P October 24 rainfall, 27\_12\_ARM is Water storage in the December 27th, 30\_11\_P November 30th Precipitation, 28\_12\_ARM is December 28th soil water storage, 20\_12\_TMED December 20th average temperature, 20\_12\_ETP December 20th potential evapotranspiration, and 25\_12\_ARM is water in the soil of the 25th of December

absolute error (MAE), and the precision was defined by the adjusted coefficient of determination ( $R^2_{ADJ}$ ) (Eqs. 1, 2, 3, and 4).

$$MSE = \frac{\sum_{i=1}^N (Y_{obs_i} - Y_{est_i})^2}{N} \quad (1)$$

$$MAPE = \frac{\sum_{i=1}^N \left( \left| \frac{Y_{est_i} - Y_{obs_i}}{Y_{obs_i}} \right| * 100 \right)}{N} \quad (2)$$

$$MAE = \frac{\sum_{i=1}^N |Y_{est_i} - Y_{obs_i}|}{N} \quad (3)$$

$$R^2_{ajustado} = \left[ 1 - \frac{(1 - R^2) \times (N - 1)}{N - k - 1} \right] \quad (4)$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^N (Y_{obs} - Y_{est})^2}{N}} \quad (5)$$

where  $Y_{obs}$  is the observed yield (bags  $ha^{-1}$ ),  $Y_{est}$  is the estimated yield (bags  $ha^{-1}$ ),  $N$  is the number of observed yield data,  $k$  is the number of FEATURES in the input layer and  $Y$  is the average estimated yield (bags  $ha^{-1}$ ).

The climate data collected from NASA-POWER together with the annual yields published by Aprosoja-MS were tabulated for the years 2002 to 2021. At first, Pearson's correlations were performed between the variables, generating Table 2. Multiple linear regression (MLR), multilayer perceptron (MLP), support vector machine (SVM), random forest (RF), extreme gradient boosting (XGBOOSTING), and gradient boosted decision (GradBOOSTING) models were calibrated with 70% and tested with 30% of available data (Schaafsma and Vark 1979). The tendency, precision, and accuracy of the calibration and testing were calculated by MAPE,  $R^2$ , RMSE, MSE, and MAE. Actual and predicted yields were interpolated into maps of spatial variation (Fig. 4).

## 2.5 Geostatistical analysis

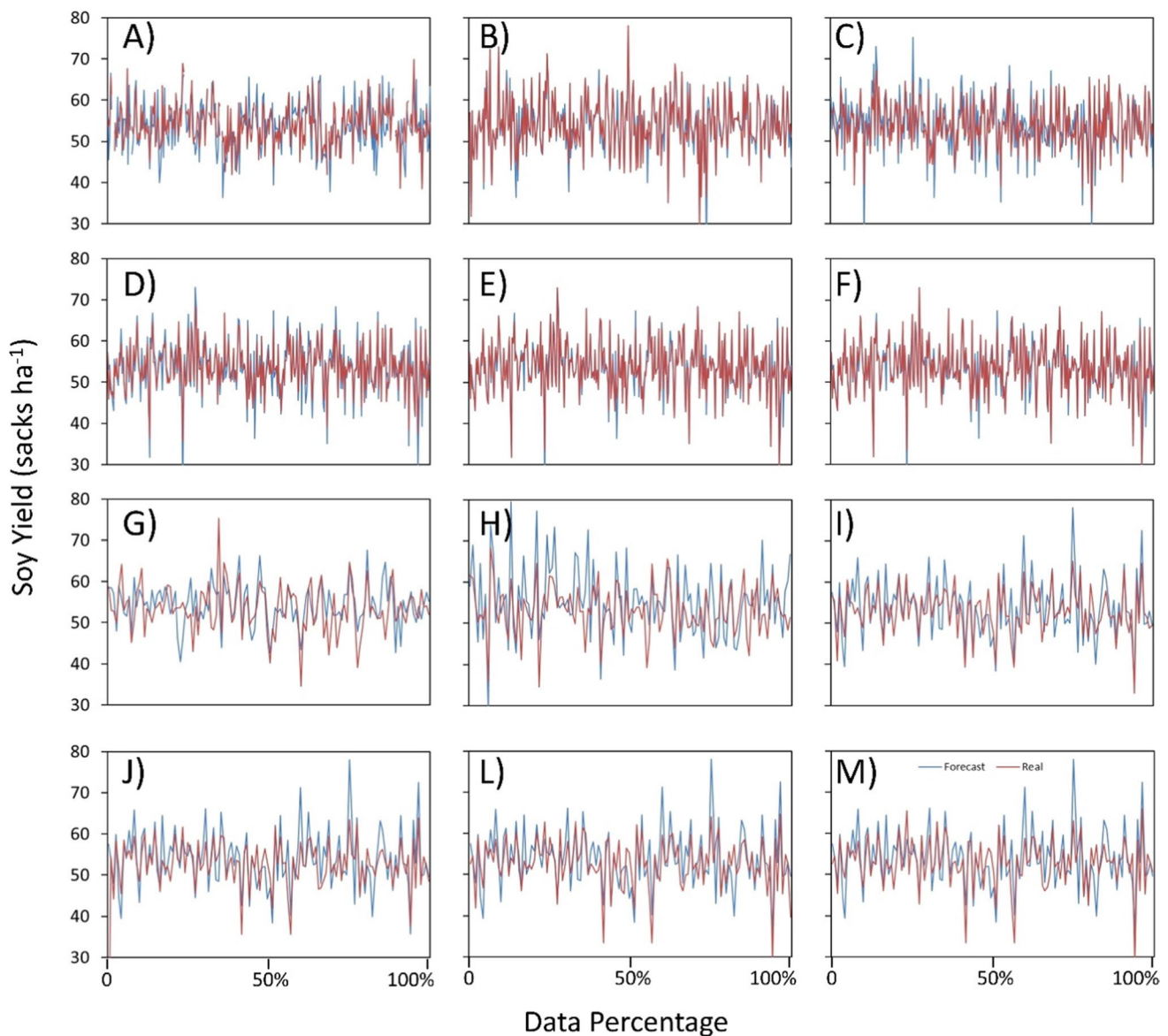
The result of the most accurate ML model was spatialized for all MS regions (Fig. 1). The spatial interpolation of actual and predicted soybean yields was generated using the geographic information system (GIS) by the kriging method (Krige 1951), using a spherical model with one neighbor and a spatial resolution of 26 km.

## 3 Results and discussion

Climate variables showed high spatial and seasonal variability throughout Mato Grosso do Sul (MS). For instance, the variable air temperature ( $T_{ar}$ ) ranged from 20.2 to 30.0 °C for the south and north of MS, respectively (Fig. 5a). These results were already confirmed by other authors (Alvares et al. 2013; Aparecido et al. 2020). Regarding seasonal variability, the lowest and highest T occurred in June and January, respectively (Fig. 6a). Corumbá is a predominant location in this region of higher T in MS. The spatial distribution showed that the highest water storage (ARM) values occurred from the center to the south, regions with ARM values higher than 360 mm per year and a great climate potential for soybean production, with the largest productions in the entire state (CONAB 2021) (Fig. 6).

The mean soybean production in MS was 53.71 ( $\pm 1.951$ ) bags  $ha^{-1}$ , with high variability between locations in MS, as observed by other authors (CONAB 2021; Pesqueira et al. 2016). Novo Mundo was the location with the lowest mean production from 2010 to 2021, with a value of 44.76 ( $\pm 8.82$ ) bags  $ha^{-1}$ , and the location with the highest mean reached 60.30 ( $\pm 6.10$ ) bags  $ha^{-1}$  (Fig. 7). Importantly, Douradina, Aral Moreira, and Laguna Carapã presented the highest





**Fig. 10** Relationship between the actual and predicted data by the algorithms in the calibration of the soybean forecast in Mato Grosso do Sul. **A** Multiple linear regression (MLR), **B** multilayer perceptron (MLP), **C** support vector machine (SVM), **D** random forest (RF), **E** extreme gradient boosting (XGBOOSTING), and **F** gradient boosted decision (GradBOOSTING). Relationship between actual and pre-

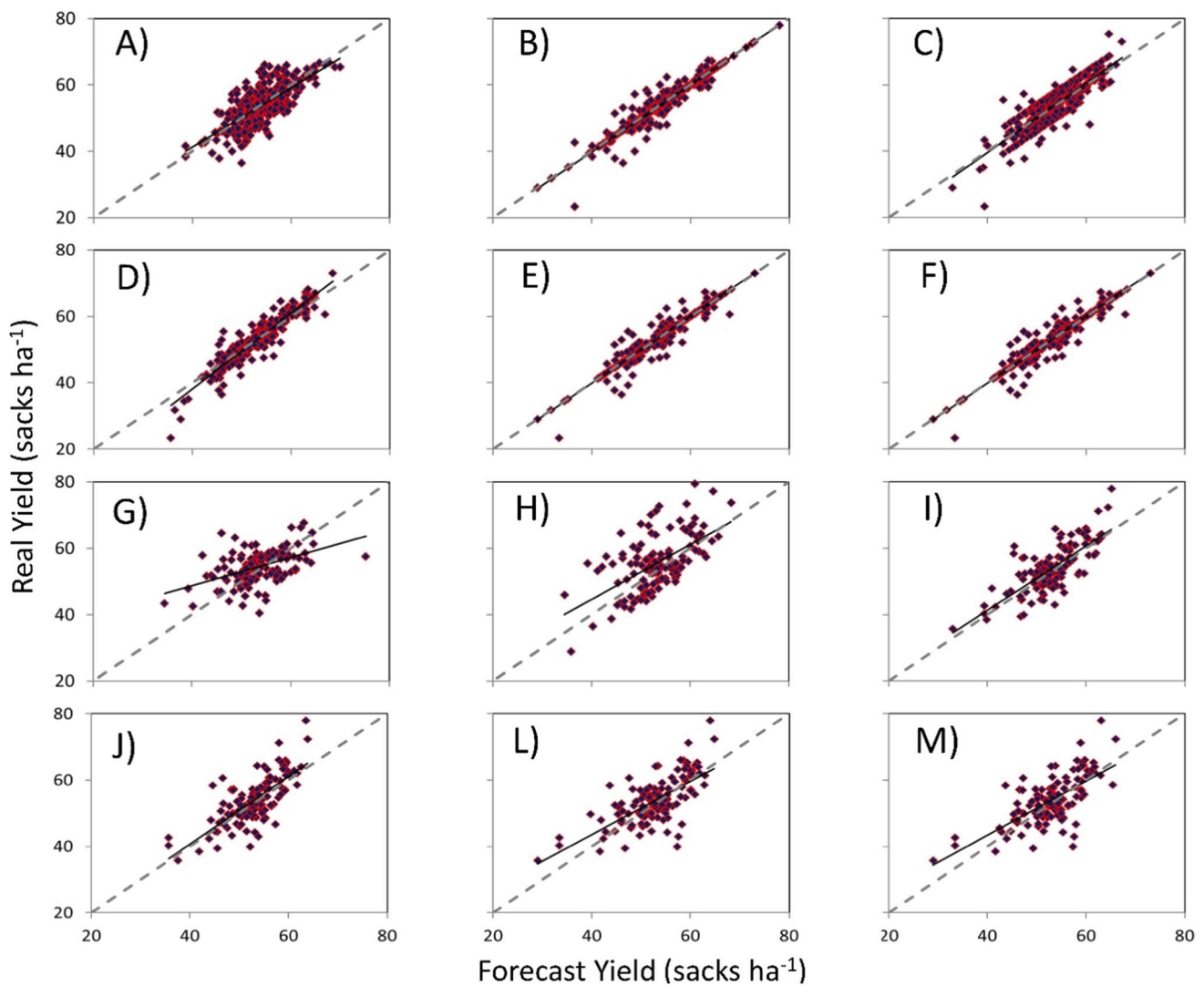
dicted data by the algorithms in the Mato Grosso do Sul soybean prediction test. **G** Multiple linear regression (MLR), **H** multilayer perceptron (MLP), **I** support vector machine (SVM), **J** random forest (RF), **L** extreme gradient boosting (XGBOOSTING), and **M** gradient boosting (GradBOOSTING)

yield variability from 2010 to 2021, also being usually the locations that are most influenced by the climate regarding the studied crop.

The temporal evolution of soybean yield showed a clear increase for all cities from 2016 onwards (Fig. 8). Cassilândia, Douradina, and Chapadão do Sul showed the highest growth, with a variation from 2010 to 2021 of 17.92, 17.80, and 16.25 bags  $\text{ha}^{-1}$ , respectively. On the other hand, Anaurilândia, Camapuã, and Batayporã showed a decrease in

yield from 2010 to 2020, with values of 6.35, 4.59, and 1.21 bags  $\text{ha}^{-1}$ , respectively. The lowest yield was observed in Douradina in 2013, followed by Mundo Novo in 2019 and Aral Moreira in 2011, with values of 23.41, 29.00, and 31.80 bags  $\text{ha}^{-1}$ , respectively. In contrast, the highest soybean yield in MS was observed in 2020 in Cassilândia, followed by Chapadão do Sul and São Gabriel do Oeste, with values of 78.02, 75.27, and 72.98 bags  $\text{ha}^{-1}$ , respectively.

Pearson's univariate correlations between soybean yield and climate variables during the phenological period showed



**Fig. 11** Performance between the real and predicted data by the algorithms in the calibration of the soybean forecast in Mato Grosso do Sul. **A** Multiple linear regression (MLR), **B** multilayer perceptron (MLP), **C** support vector machine (SVM), **D** random forest (RF), **E** extreme gradient boosting (XGBOOSTING), and **F** gradient boosting (GradBOOSTING). Dashed line: 1:1 line. Performance between

actual and predicted data by the algorithms in the Mato Grosso do Sul soybean prediction test. **G** Multiple linear regression (MLR), **H** multilayer perceptron (MLP), **I** support vector machine (SVM), **J** random forest (RF), **L** extreme gradient boosting (XGBOOSTING), and **M** gradient boosting (GradBOOSTING). Dashed line: 1:1 line

distinct relationships and different intensities (Table 2). For instance, soil water storage (ARM) showed negative, neutral, and positive correlations in October, November,

and December, respectively. It demonstrates that the same climate element has different influences depending on the phenological period in which the plant is at that moment.

**Table 3** Static indices from the calibration and testing of machine learning algorithms to predict soybean yields in Mato Grosso do Sul, Brazil

| Algoritmos   | Calibration |       |      |       |      | Test  |       |      |       |      |
|--------------|-------------|-------|------|-------|------|-------|-------|------|-------|------|
|              | MAPE        | R2adj | RMSE | MSE   | MAE  | MAPE  | R2adj | RMSE | MSE   | MAE  |
| MLR          | 6.01        | 0.60  | 4.38 | 19.17 | 3.13 | 9.33  | 0     | 6.60 | 43.51 | 5.07 |
| MLP          | 2.00        | 0.91  | 2.07 | 4.28  | 0.98 | 12.46 | 0.09  | 9.70 | 94.00 | 6.85 |
| SVM          | 5.65        | 0.72  | 3.55 | 12.62 | 2.92 | 6.91  | 0.48  | 4.79 | 22.96 | 3.68 |
| RF           | 3.55        | 0.92  | 2.57 | 6.58  | 1.79 | 6.57  | 0.71  | 4.79 | 22.99 | 3.53 |
| XGBOOSTING   | 1.84        | 0.95  | 2.06 | 4.24  | 0.92 | 7.53  | 0.62  | 5.37 | 28.83 | 3.97 |
| GradBOOSTING | 1.98        | 0.95  | 2.12 | 4.49  | 0.99 | 7.51  | 0.62  | 5.37 | 28.89 | 3.96 |

Water deficit showed an opposite behavior to that of ARM, as it had a positive correlation in October and a negative correlation in December. It evidences that DEF promotes an increase in soybean yield at the beginning of maturation (Souza et al. 2013; Zhang et al. 2007).

The 15 highest correlations were analyzed separately due to the large number of values found (Table 2), aiming at a more thorough evaluation (Fig. 9). Overall, the highest indirect correlations (–) were observed for 2\_10\_TDEW

(dew point temperature on the second day of October), 21\_12\_TMED (mean air temperature on the twenty-first day of December), and 21\_12\_TMAX (maximum air temperature on the twenty-first day of December), with values of –0.41, –0.40, and 0.38 respectively. Importantly, all these variables measure the kinetic energy of the environment and prove that an increase in the respective temperatures at these times promotes a significant reduction in soybean yield in Mato Grosso do Sul.

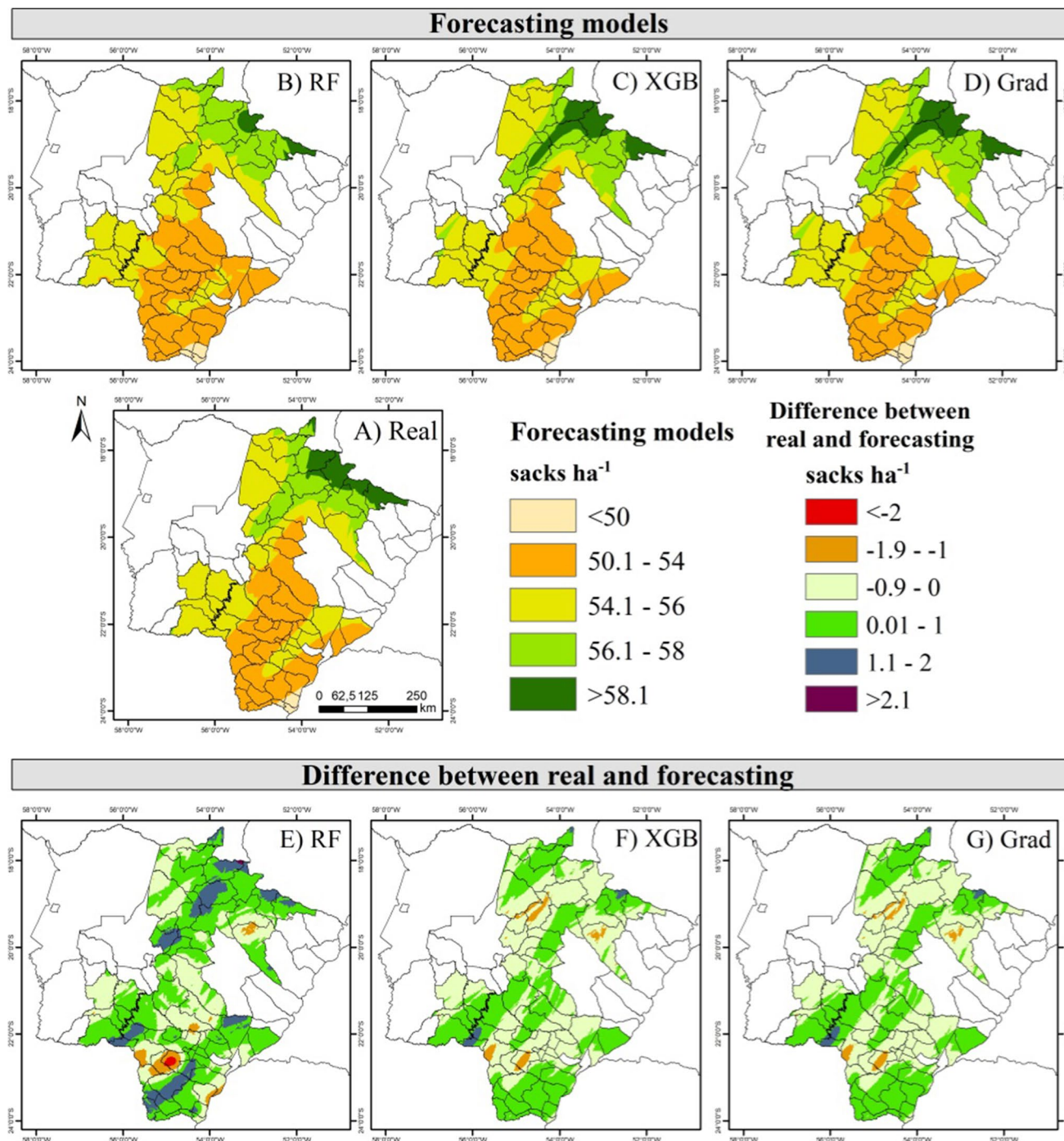


Fig. 12 Spatial variation of actual soybean yield data and data predicted by machine learning algorithms using weather



In contrast, the highest direct correlations (+) were observed for 26\_12\_ARM and 24\_10\_P, with values of 0.36 and 0.35, respectively. It shows that an increase in soil water storage and precipitation during these periods promotes higher soybean crop yields (Adeboye et al. 2017; Souza et al. 2013). It occurs because plants will have more water supply to carry out the photosynthesis process and, consequently, store more carbohydrates, lipids, and proteins in the soybean grain.

The calibrated ML algorithms showed high precision and accuracy in both calibration and testing (Table 3). For instance, the best model in the calibration was XGBOOSTING, which showed MAPE,  $R^2$ , RMSE, MSE, and MAE values of 1.84%, 0.95, 2.06%, 4.24%, and 0.921%, respectively. MLR was the algorithm with the lowest accuracy, showing MAPE,  $R^2$ , RMSE, MSE, and MAE values of 6.01%, 0.60, 4.38%, 19.17%, and 3.13%, respectively. MLR is one of the simplest methods and very used. However, when the data set of independent variables presents a greater quantity of samples and/or has multicollinearity, the method is not successful and presents deviations (Carvalho Júnior et al. 2016).

The machine learning algorithms used to predict soybean yield continue with good, precise, and accurate results in tests with independent data. For instance, the best model in the test was the Random Forest (RF), which presented MAPE,  $R^2$ , RMSE, MSE, and MAE values of 6.57%, 0.71, 4.79%, 22.99%, and 3.53%, respectively. This better performance of RF is explained by Breiman (2001) and Hoffman et al. (2020).

On the other hand, MLR was the algorithm with the lowest accuracy, with MAPE,  $R^2$ , RMSE, MSE, and MAE values of 9.33%, 0.00%, 6.60%, 43.52%, and 5.07%, respectively. The relationship between time and performance of each algorithm is shown in Figs. 10 and 11. This low performance of MLR has also been observed by other authors (Fishman et al. 2010; Kaul et al. 2005).

Figure 12 shows the maps with the actual yield (Fig. 12A), predictions of the three best algorithms, that is, RF (Fig. 11B), XGB (Fig. 12C), and Grad (Fig. 11D), and deviations between the algorithms and actual yield (Fig. 12E, F, and G). The highest yields occurred in the north/northeast, mainly in Alcínópolis, Costa Rica, and Chapadão do Sul, while the lowest soybean yields were observed in the center/south of Mato Grosso do Sul (Fig. 12A). The three best algorithms followed this spatial variability that soybean yield has in the state of Mato Grosso do Sul.

The spatial variation of deviations of the machine learning algorithms occurred heterogeneously throughout the state of Mato Grosso do Sul (Fig. 11E, F, and G) and ranging from  $-4.3$  to  $4.1$  bags  $\text{ha}^{-1}$  (Fig. 10). The deviations

of the RF algorithm were more dispersed and high, with a higher presence of values between 1.1 and 2.0 bags  $\text{ha}^{-1}$  in all regions evaluated in the state of Mato Grosso do Sul (Fig. 12E). Moreover, the deviations of XGB and Grad algorithms were very similar in all evaluated regions (Fig. 12F, G).

## 4 Conclusions

- Soybean yield can be accurately predicted in Mato Grosso do Sul using machine learning algorithms and climate.

Climate elements with the highest positive and negative correlations were 26\_12\_ARM (storage on the twenty-sixth day of December) and 2\_10\_TDEW (dew point temperature on the second day of October), occurring in the phenological periods of grain filling and emergence, respectively.

- Multiple linear regression was the algorithm with the lowest accuracy and low precision for predicting soybean yield using climate, showing MAPE values of 6.01 and 9.33 in the calibration and testing, respectively.

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**Author contribution** Guilherme B Torsoni: formal analysis, investigation, data curation, writing—original draft, writing—review and editing, visualization. Lucas Eduardo de Oliveira Aparecido: conceptualization, methodology, 562 supervision, project administration. Gabriela Marins dos Santos: writing—review and editing. Alisson Gaspar Chiquitto: writing—review and editing. Glaucio de Souza Rolim: writing—review and editing. José Reinaldo da Silva Cabral de Moraes: writing—review and editing. All authors read and approved the final manuscript.

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**Availability of data and material** The data/ material is opened.

**Code availability** The software used was python and scripts are available.

## Declarations

**Competing interests** The authors declare no competing interests.

**Ethics approval** Not applicable.

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